

Reflection Journal

For this final project, I had to use the same Rajasthan Heatwave dataset from the midterm. This time since it was a final, we had to do it individually instead of a group, so I did end up taking help from AI a lot for the coding intensive part, but I put AI disclaimers and the prompt used. The first step was going from 70/30 train/test split to a 70/15/15 train/validation/test split without redoing the preprocessing from steps. I instead of reloading the raw CSV and running through fillna, encoding, and scaling all over again, I just split the 30% test set in half, which gave me a 15% validation set and a 15% test set, and both kept the dataset's real imbalance since only the training data can ever be touched by SMOTE. This is actually the more correct way to do it since I wasn't reprocessing anything that had already been validated. This wasn't because I was being lazy and nor that it was a shortcut

After training all six models, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, KNN, and SVC. Three of them, Decision Tree, Random Forest, and Gradient Boosting, came back at 100% on every single metric. Accuracy, precision, recall, F1, ROC-AUC, all were 100%. It was obvious that a perfect score across three different models is rarely if ever possible, so I checked Random Forest's feature importances and found that TMAX, TMIN, and TEMP2M made up about 79% of what the model was using to make decisions. Heatwaves are defined by temperature crossing a cutoff, so the model wasn't technically predicting anything, it was just reading the answer off the temperature columns.

I dropped those three columns and retrained everything, and the scores came back down to something more reasonable, around 92 to 97% accuracy with F1-scores between 0.51 and 0.70. Random Forest ended up the best of the six on its own, and the Voting and Bayesian ensembles

were easy to build after that since I already had my top 3 models from the validation results.

Weighting the Bayesian ensemble by F1-score barely changed anything compared to just averaging normally, the weights came out close to equal since Random Forest, SVC, and Gradient Boosting weren't that far apart anyway.

Doing this alone instead of with my group was actually much better for me personally because I was able to work at my pace and actually be aware of why, what, and how something happened instead of trying to ask my group member that for their parts.

Figuring out the leak was probably the most important part of the whole project. It's would have been great to see a 100% and move on, but I knew from our lessons it's close to impossible.

Random Forest ended up being my strongest model, and the ensembles traded some precision for better recall, so which one's actually to use depends on whether missing a heatwave or a false alarm matters more.